

**CENTER FOR DRUG EVALUATION AND
RESEARCH**

APPLICATION NUMBER:

202292Orig1s000

**CLINICAL PHARMACOLOGY AND
BIOPHARMACEUTICS REVIEW(S)**

CLINICAL PHARMACOLOGY REVIEW

NDA: 202292

Submission Date: 5 DEC 2011

Submission Type; Code: NME
Brand/Code Name: *Pending*
Generic Name: Crofelemer
Primary Reviewer: Kristina Estes, PharmD
Secondary Reviewer: Sue Chih Lee, PhD
OCP Division: Division of Clinical Pharmacology 3
OND Division: Division of Gastroenterology and Inborn Errors Products
Sponsor: Salix
Relevant IND(s): 51,818; (b) (4)
Formulation; Strength(s): 125 mg Tablet
Proposed Indication: (b) (4) diarrhea in HIV⁺ patients on antiretrovirals
Proposed Dosage
Regimen: 125 mg twice daily

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1 Executive Summary

Crofelemer is a botanical product intended for the treatment of (b) (4) diarrhea in HIV⁺ patients taking antiretroviral (ARV) therapy. There are no other drugs that have been approved for this indication. Other antidiarrheal medications (ADMs), including loperamide, methylcellulose, and bismuth subsalicylate, are used to manage diarrhea but have limited effect in patients with chronic secretory diarrhea. Chronic diarrhea is of particular concern in patients with HIV due to concerns about ARV compliance and the potential for subsequent

loss of clinical effect. The sponsor seeks to market this botanical drug as a 125 mg enteric-coated tablet in the U.S.

This NDA is submitted under the provisions of 505(b)(1). To support the clinical pharmacology section of this NDA, the sponsor has submitted the results of nine Phase 1 studies, a Phase 2 study, and two Phase 3 trials. The Phase 1 studies in healthy volunteers included four single- and multiple-dose pharmacokinetic (PK) studies, three studies of the maximum tolerated dose, one drug-drug interaction study, and one food effect study. The sponsor has also submitted some data from several in vitro studies of crofelemer that addressed the following: plasma protein binding, Caco-2 permeability, CYP inhibitory potential, and transporter inhibitory potential.

1.1 Recommendations

From the viewpoint of the Office of Clinical Pharmacology, the Clinical Pharmacology and Biopharmaceutics information in the NDA is acceptable provided that mutual agreement on label language can be reached between the sponsor and the Agency.

1.2 Post-Marketing Commitments

1. The Office of Clinical Pharmacology recommends the sponsor perform an in vitro study to determine whether crofelemer is an inhibitor of P-glycoprotein (p-gp) and BCRP, transporters expressed in the gut. The sponsor has conducted one in vitro study of p-gp but it was performed with (b) (4) crofelemer. This study would be a post-marketing commitment. Depending on the results, further studies may be necessary.

2. We also recommend an in vivo drug interaction study to characterize the potential of crofelemer to inhibit CYP3A4 at the level of the gut using a sensitive 3A4 substrate such as midazolam or a suitable antiretroviral drug because in vitro studies suggested the potential for such in vivo interactions. The study would also be a post-marketing commitment.

1.3 Summary of CPB Findings

Pharmacokinetics in Healthy Volunteers and Patients

The systemic exposure of crofelemer was low in healthy volunteers and patients; therefore, the PK was not able to be fully characterized. Crofelemer, a proanthocyanidin, is described by the sponsor as a polymer and is made up of linear chains of catechin, epicatechin, gallocatechin, or epigallocatechin monomer units. These chains reportedly range from 1 to 28 units, with an average length of 5 to 7 units. Data from the literature confirms that proanthocyanidin polymerization greatly impairs intestinal absorption [AJCN 2005;81:230s-42s]. Results from the food effect study, which included rich sampling and an LC/MS/MS assay with an LOQ of 50 ng/mL, showed only two samples with detectable crofelemer (54.7 and 70.7 ng/mL). Trough blood samples collected in the ADVENT trial and also analyzed using the same assay method detected crofelemer in only 15 of 456 samples with a maximum concentration of 77 ng/mL.

Metabolism

The metabolism of crofelemer, in vitro or in vivo, was not determined by the sponsor. Data from the literature suggests gut microflora are responsible for degradation of proanthocyanidins; however, data from crofelemer is not available in humans [AJCN 2005;81:230s-42s]. Due to the complexity of the molecules present in this drug, no further studies will be requested of the sponsor at this time.

Drug Drug Interactions

The results of in vitro studies suggest that crofelemer has the potential to inhibit CYP 3A4 at the concentration expected in the gut. Inhibition of other CYPs is also possible, however, only 3A4 is widely expressed in the gut. Inhibition of CYP enzymes in the liver is unlikely

given systemic exposure to crofelemer is very low. Crofelemer does not induce expression of CYPs 1A2, 2B6, or 3A in human hepatocytes.

The potential for crofelemer to interact with ARVs was explored in study 37554-103. Crofelemer (500 mg Q6H) was administered to 28 healthy volunteers for five days. Three ARVs, nelfinavir, zidovudine, and lamivudine, were administered as a single dose together on Day 5. The results of the study showed crofelemer had no effect on the exposure of zidovudine and nelfinavir; however, there was a 20% decrease in lamivudine exposure. However, this interaction with lamivudine was not deemed to be clinically important even under conditions of use in standard clinical practice.

Dose-Selection

The dose-response for crofelemer on stool weight and clinical response was determined in the first clinical trial (37554-210) and the first phase of ADVENT, respectively. The doses selected for 37554-210 were 250 mg and 500 mg four times daily and placebo. The doses selected for that ADVENT were 125 mg, 250 mg, and 500 mg twice daily for 4 weeks (primary endpoint) versus placebo, then for an additional 20 weeks in which all placebo patients randomized to one of the three crofelemer doses for the remainder of the time. In study 37554-210, there was no clear dose-response and no consistent response relative to placebo. In ADVENT, there was no apparent dose response among the three dose groups but all three were superior to placebo with regard to the primary endpoint. Multiple dose levels were not included in the Phase 2 study (37554-209), which assessed stool weight following a 4-day treatment with 500 mg crofelemer administered four times daily. It should be noted that only one study (ADVENT) supports the application at the proposed dose. The other studies used different endpoints, dosing regimens, treatment duration, and crofelemer formulations.

2 QBR

2.1 Background

2.1.1 What regulatory background or history information contributes to the assessment of the clinical pharmacology and biopharmaceutics of this drug?

(b) (4)
(u) (4)
(IND 51,818 filed in 1996). Studies in patients with chronic HIV-associated diarrhea were initiated in 1997 and reviewed by the Division of Antiviral Drug Products. The EOP2 meeting was held in 1998 and the Agency agreed that an NDA submission with a single Phase 3 trial would be acceptable if robust. Study 37554-210, a Phase 3 trial in AIDS patients was completed in 1998; however, in the subsequent pre-NDA meeting, the sponsor was informed that the safety database was insufficient and an additional Phase 3 trial would be required. A new sponsor met with the Division of Gastroenterology Products in 2004 to discuss the design of a new clinical trial (ADVENT), which was initiated in 2007 and completed in 2011.

There are no approved drugs in the U.S. for the treatment of diarrhea in patients with HIV on antiretroviral therapy. Diarrhea contributes to poor compliance with antiretroviral regimens, which may increase the emergence of treatment-resistant HIV and disease progression. The Agency concurs that diarrhea in HIV+ patients is serious and potentially life-threatening; therefore, the Agency designated crofelemer for Fast Track development and has agreed to a priority review.

2.1.2 List the in vitro and in vivo clinical pharmacology and biopharmaceutics studies and the clinical studies with PK and/or PD information submitted in the NDA or BLA.

Nine clinical pharmacology studies were performed in healthy volunteers and three studies were performed in HIV+ patients. The following table lists the clinical studies that were submitted in support of this application.

Table 1. Summary of Clinical Pharmacology Studies

Study Identifier	Type of Study	Number of subjects	Duration of treatment	Patient Population
SP-303-I-01	Single-dose PK	14	1 day	Healthy
SP-303-I-02	Maximum tolerated dose	37	1 day	Healthy
SP-303-I-03	Maximum tolerated dose	24	7 days	Healthy
SP-303-I-04	Maximum tolerated dose	16	7 days	Healthy
SP-303-I-07	Multiple-dose PK	5	3.5 days	Healthy
37554-103	Drug Interaction	28	5 days	Healthy
51818-101	Single-dose PK	6	1 day	Healthy
51818-102	Single- and multiple-dose PK	8	2 days	Healthy
CFFE1091	Food Effect	28	1 day	Healthy
NP-303-101 (ADVENT)	Phase 3	376	24 weeks	HIV ⁺
37554-209	Phase 2	85	4 days	HIV ⁺
37554-210	Phase 3	400	27 days	HIV ⁺

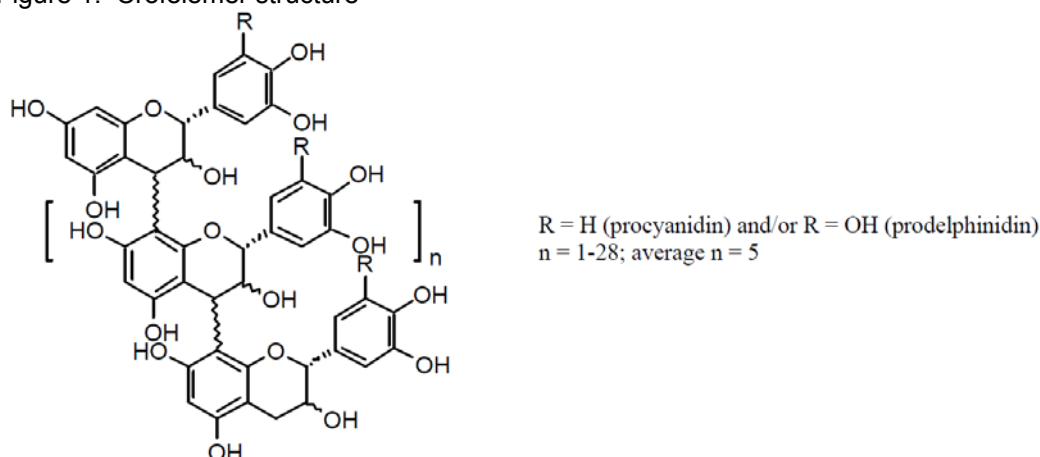
In addition, the following in vitro studies were conducted in support of the application: 1) intestinal permeability using Caco-2 cells; 2) protein binding in human plasma; 3) effect of crofelemer on the metabolism of probe substrates for CYP 1A2, 2B6, 2C9, 2C19, 2D6, 2E1 and 3A4 isozymes in pooled human liver microsomes; and 4) interaction with transporters including MRP2, MRP4, P-gp, OATP2B1, PEPT1, OATP1A2, and ASBT.

2.2 General Attributes of the Drug

2.2.1 What are the highlights of the chemistry and physical-chemical properties of the drug substance, and the formulation of the drug product?

Crofelemer is classified as a proanthocyanidin (or condensed tannin), which are ubiquitous naturally-occurring phenolic substances found in a wide variety of plants. The drug substance is extracted from the bark latex of the plant *Croton lechleri* in the western Amazon regions of South America. Crofelemer is described as a polymer with linear chains of catechin, epicatechin, gallocatechin, or epigallocatechin monomer units. These chains of monomers range from one to 28 units with an average of five to seven units reported. The average molecular weight is estimated to be 1700 to 2500 Dalton. Crofelemer is highly water soluble over a wide pH range but exhibits very poor permeability in vitro.

Figure 1. Crofelemer structure



Crofelemer is formulated as a 125 mg enteric coated tablet. The sponsor indicates that early in the development they assumed the drug substance needed to be protected from acid so it could be delivered intact to the intestine, where it exerts a local effect. However, a study of crofelemer under gastric conditions demonstrated the stability of crofelemer in acid for up to 24 hours. The sponsor elected to continue manufacturing an enteric coated tablet in order to be consistent with the product that was used in the clinical trial. See review by Dr. Seggel of ONDQA for a complete discussion of this issue.

A variety of formulations were used in the clinical pharmacology studies and the clinical trials (see table below). In addition, several different manufacturers were responsible for drug production, coating, and encapsulation throughout clinical development. The to-be-marketed formulation was used in the ADVENT clinical trial and in the food effect study (CFE1091). The early clinical trial (study 37554-210) used both an encapsulated tablet and beads in capsule. The Phase 2 study (37554-209) used beads in capsule.

Table 2. Summary of formulations used in clinical studies

Formulation	Study Identifier	Lot Number	Manufacturer
Oral solution (powder dissolved in water or 0.2% xanthan gum)	SP-3-3-I-01	SH-04-06211	(b) (4)
	SP-3-3-I-02	SH-05-03032	
	SP-3-3-I-03	SH-06-07152	
	SP-3-3-I-04	PL-018	
	SP-3-3-I-07	PL-022	
Beads in capsule	51818-101	960056A	
Beads in capsule	37554-209	970031, 970147	
	37554-210	980030A, B; 980079A, B	
Encapsulated tablet	37554-210	980038A, 980062A	
Encapsulated tablet	37554-103	980080	
Tablet	51818-102	970096A, 970096B	
Tablet	NP303-101 (ADVENT)	3059847R, 3061308R, 3062741R, 3063506R, 3064439R, 3065505R, 3067354R, 3067355R, 3072911R, 3072912R, 3061703R, 3062743R, 3063507R, 3064440R, 3065503R, 3067356R	

	CFFE1091	3075208	
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2.2.2 What are the proposed mechanism of action and therapeutic indication(s)?

Crofelemer has been shown to be a potent inhibitor of the cystic fibrosis transmembrane regulator conductance (CFTR) and calcium-activated chloride (CaCC) channels *in vitro*. These two apical chloride transporters regulate chloride ion secretion in enterocytes. Intestinal fluid secretion involved Cl⁻ influx into enterocytes through transporters on the basolateral side and then efflux through apical chloride channels. Sodium and water follow passively. Inhibition of Cl⁻ efflux by crofelemer results in an antidiarrheal effect by decreasing water flow in the lumen of the gastrointestinal tract. The proposed indication is *the control and symptomatic relief of diarrhea in patients with HIV/AIDS on antiretroviral therapy*.

2.2.3 What are the proposed dosage and route of administration?

The proposed dose is one 125 mg crofelemer tablet by mouth twice daily.

2.2.4 What drugs with the same indication are approved in the U.S.?

There are currently no drugs approved for the treatment of diarrhea in patients with HIV or AIDS on antiretroviral therapy.

2.3 General Clinical Pharmacology

2.3.1 What are the design features of the clinical pharmacology and biopharmaceutics studies and the clinical studies used to support dosing or claims?

Very little PK data was generated from the clinical pharmacology studies and the clinical trials due to the low bioavailability of crofelemer. Three studies were conducted to assess the maximum tolerated dose (dose range: 3-20 mg/kg and 150-2000 mg), which found no dose-response relationship for safety. One phase 2 study and two phase 3 trials were also conducted to support claims.

Phase 2 Study

Study 37554-209 was conducted in HIV+ patients to assess the effect of crofelemer 500 mg Q6H on stool weight following four days of treatment. There was a significantly greater decrease in stool weight from baseline to Day 4 in the crofelemer group compared to placebo.

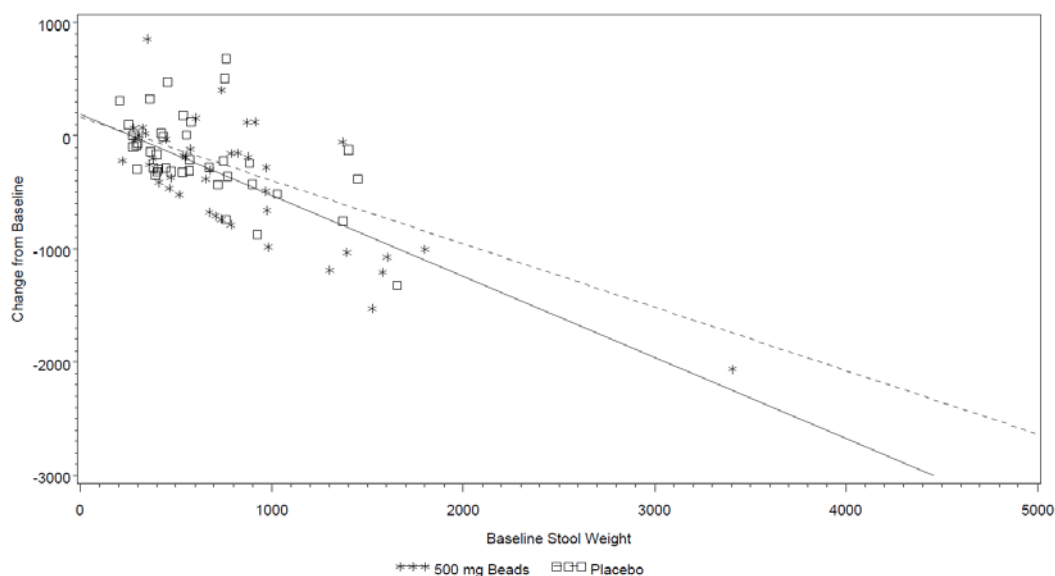
Table 3. Stool weight in the Phase 2 study at baseline and Day 4

Inpatient Period	Crofelemer 500 mg beads n=43	Placebo n=42
Stool Weight at Baseline (g)		
Mean (SD)	861.3 (604.67)	730.9 (720.14)
Median	707.7	547.0
Min, Max	220, 3407	206, 4701
P-value (vs. placebo) ^{a,b}	0.3074	--
Categories of stool weight at baseline, n (%)		
Low (≤ 740 g)	24 (55.8)	28 (66.7)
High (> 740 g)	19 (44.2)	14 (33.3)
P-value (vs. placebo) ^{a,b}	0.2725	--
Change: Baseline to Day 4 (g)		
Mean (SD)	-401.3 (531.65)	-192.4 (381.57)
Median	-267.5	-232.8
Min, Max	-1815, 854	-1319, 683
P-value (vs. placebo) ^b	0.0335	--

Abbreviations: SD=standard deviation, min=minimum, max=maximum.

The mean and median stool weight at baseline was higher in the crofelemer group than the placebo group. The sponsor indicated that the decrease in stool weight appeared to be greater in patients with a baseline stool weight > 740 g compared to those ≤ 740 g.

Figure 2. Change in stool weight from baseline to Day 4 (ITT population)



This study also included daily measurements of stool chloride concentrations and stool Cl^- output. Crofelemer treated subjects had a statistically significant reduction in stool Cl^- concentrations and stool Cl^- output from baseline to Day 4 relative to placebo.

Table 4. Stool chloride output change from baseline to Day 4

Stool Chloride Output (mg)	Placebo (N = 25)	500 mg Beads (N = 26)
Baseline Stool Chloride Output		
n	25	26
Mean (SD)	1386.5 (4606.50)	854.7 (1148.24)
Median	353.9	458.5
Min, Max	64, 23412	70, 4789
P-Value (vs. Placebo)		0.5916
Baseline Stool Chloride Output - n (%)		
Low (≤ 500)	17 (68.0%)	16 (61.5%)
High (> 500)	8 (32.0%)	10 (38.5%)
P-Value (vs. Placebo)		0.6033

Change from Baseline to Day 4		
n	24	26
Mean (SD)	-72.6 (400.18)	-532.4 (972.02)
Median	-113.8	-194.2
Min, Max	-839, 956	-3145, 525
Least Square Mean (Std. Err.)	-116.8 (161.50)	-650.6 (142.22)
Estimate (vs. Placebo)		-533.8
P-Value (vs. Placebo)		0.0136
Estimate (vs. Placebo at Baseline Low)		-133.9
P-Value (vs. Placebo at Baseline Low)		0.5834
Estimate (vs. Placebo at Baseline High)		-933.7
P-Value (vs. Placebo at Baseline High)		0.0118

Overall, there was a significant change in stool Cl⁻ output following four days of treatment. The difference is most apparent for patients with a high Cl⁻ output at baseline. The difference is not significant for patients with a low Cl⁻ output at baseline, which is consistent with the proposed mechanism of action.

Table 5. Stool chloride concentration change from baseline to Day 4

Stool Chloride Concentration (mg/gm)	Placebo (N = 25)	500 mg Beads (N = 26)
Baseline Stool Chloride Concentration		
n	25	25
Mean (SD)	0.852 (0.9098)	0.747 (0.6310)
Median	0.673	0.520
Min, Max	0.22, 4.91	0.21, 2.60
P-Value (vs. Placebo)		0.5973
Baseline Stool Chloride Output - n (%)		
Low (<= 0.6)	12 (48.0%)	15 (60.0%)
High (> 0.6)	13 (52.0%)	10 (40.0%)
P-Value (vs. Placebo)		0.3605
Change from Baseline to Day 4		
n	24	25
Mean (SD)	0.123 (0.7138)	-0.245 (0.5556)
Median	0.057	-0.093
Min, Max	-1.29, 1.97	-1.51, 0.74
Least Square Mean (Std. Err.)	0.091 (0.1032)	-0.371 (0.1052)
Estimate (vs. Placebo)		-0.462
P-Value (vs. Placebo)		0.0024
Estimate (vs. Placebo at Baseline Low)		0.036
P-Value (vs. Placebo at Baseline Low)		0.8619
Estimate (vs. Placebo at Baseline High)		-0.961
P-Value (vs. Placebo at Baseline High)		0.0002

Overall, there was a significant change in stool Cl⁻ concentration following four days of treatment. The difference is most apparent for patients with a high Cl⁻ output at baseline but is not apparent for patients with a low Cl⁻ output at baseline.

Phase 3 Studies

The first clinical trial (37554-210) explored the effect of three crofelemer dosing regimens (250 or 500 mg *tablet* Q6H or 500 mg *encapsulated beads* Q6H) on stool weight over a 27-day treatment period.

Table 6. Stool weight in study 37554-210 at baseline and Day 7

Inpatient Period	Crofelemer 250 mg tablets N=102	Crofelemer 500 mg tablets N=100	Crofelemer 500 mg beads N=100	Placebo N=98
Stool weight at baseline (g)				
Mean (SD)	782.1 (506.79)	901.5 (882.50)	705.3 (408.24)	749.4 (475.41)
Median	634.0	641.0	605.5	567.5
Minimum, maximum	300, 2700	193, 7347	300, 3200	317, 3068
P-value (vs. placebo) ^a	0.6886	0.0684	0.6062	-
Categories of stool weight at baseline, n (%)				
Low (≤ 1000 g)	78 (76.5)	79 (79.0)	90 (90.0)	76 (77.6)
High (> 1000 g)	24 (23.5)	21 (21.0)	10 (10.0)	22 (22.4)
P-value (vs. placebo) ^a	0.8390	0.8305	0.0202	-
Change: Baseline to Day 7 (g)				
Mean (SD)	-327.5 (399.78)	-420.1 (894.38)	-309.5 (383.16)	-332.0 (439.97)
Median	-263.0	-259.0	-305.0	-292.5
Minimum, maximum	-1807, 628	-7217, 1498	-2152, 429	-2201, 1149
Least Square Mean (SE)	-440.9 (60.82)	-688.1 (64.14)	-504.8 (85.41)	-472.2 (64.13)
P-value (vs. placebo) ^a	0.7028	0.0107	0.7469	-

Source: Table 14.2.1. Abbreviations: SD=standard deviation, SE=standard error.

There was a statistically significant decrease in stool weight in only one treatment group (500 mg *tablet* Q6H) relative to placebo. The results are likely driven by a higher mean baseline stool weight in that group relative to the others and do not suggest a dose response relationship for the primary endpoint.

The ADVENT trial was designed with a 24-week dose-ranging stage (125, 250, or 500 mg twice daily) followed by an additional 24-week stage with the lowest effective dose identified in the first phase. No difference in clinical response, defined as < 2 watery stools per week during at least 2 of 4 weeks of the placebo-controlled phase, was observed during the dose-ranging stage; therefore, the lowest dose (125 mg twice daily) was selected for the second stage of the study and is the proposed dose for marketing.

Demographics

The vast majority of patients in the Phase 2 and Phase 3 studies were males. The Phase 2 study (37554-209) had the highest percentage of males (98%) while the ADVENT trial and Study 37554-210 were composed of 85% and 89% males, respectively. The mean ages in these three studies ranged from 39 to 45 years of age. The majority of patients were White (39% to 68%) or African American (12% to 38%). By ethnicity, Hispanics accounted for 6% to 26% of the patient population in these three studies.

2.3.2 What is the basis for selecting the response endpoints, i.e., clinical or surrogate endpoints, or biomarkers (also called pharmacodynamics, PD) and how are they measured in clinical pharmacology and clinical studies?

For the ADVENT trial, the primary efficacy endpoint was clinical response, defined as ≤ 2 watery stools per week for at least 2 weeks of the 4-week placebo-controlled phase. A reduction in the number of watery stools to ≤ 2 was considered clinically significant because the diarrhea experienced by these patients can have a major impact on quality of life and may result in clinical consequences such as dehydration and electrolyte loss.

The primary efficacy endpoint in the Phase 2 (37554-209) and the first Phase 3 trial (37554-210) was decrease in stool weight from baseline. This endpoint was selected as a reasonable measure of secretory diarrhea and as proof-of-concept despite the absence of a defined clinical benefit. Secondary endpoints for both studies included change in stool frequency, change in daily GI symptom score and the measure of relief scores. The Phase 2 study also studied the change from baseline in daily stool Cl^- concentration. The inhibition of Cl^- secretion to the intestinal lumen is purported to be the mechanism of action for crofelemer.

2.3.3 Are the active moieties in the plasma (or other biological fluid) appropriately identified and measured to assess pharmacokinetic parameters and exposure response relationships?

Crofelemer is a proanthocyanidin that appears to have very low systemic absorption. The drug product consists of polymeric chains of 1-28 units, with an average length of 5-7 units. Typically, absorption is limited to monomeric and dimeric proanthocyanidins. An effort was made to fully characterize the PK of crofelemer; however, very few blood samples had measurable crofelemer. For example, in the food effect study, which used a single 500 mg dose of crofelemer, only two of 840 blood samples had crofelemer levels above the LOQ. A sparse sampling scheme was included in the clinical trials but, as with the studies in healthy volunteers, few samples contained measurable crofelemer.

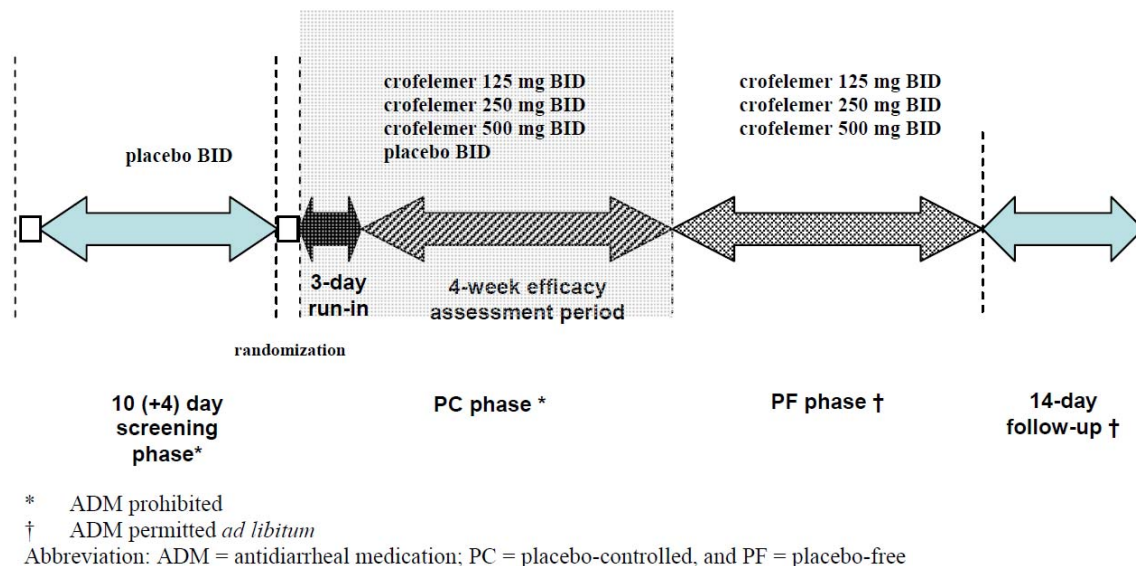
The analytical methods changes over the two decades of clinical development. HPLC was used in the majority of studies (LOQ ranged from 50 to 500 ng/mL) while LC/MS/MS was used in the ADVENT and food effect studies (LOQ 50 ng/mL). Despite use of newer analytical methods, several issues regarding the methodology remain and there are additional questions about the molecular composition of the drug substance. Information from the sponsor related to this issue is being evaluated by the CMC reviewer and may impact the need for additional studies pre- or post-approval. See 2.9.4. We further recommend the sponsor improve assay sensitivity in future studies.

2.4 Exposure-Response

2.4.1 What are the characteristics of the exposure-response relationships (dose-response, concentration-response) for efficacy? If relevant, indicate the time to the onset and offset of the pharmacological response or clinical endpoint.

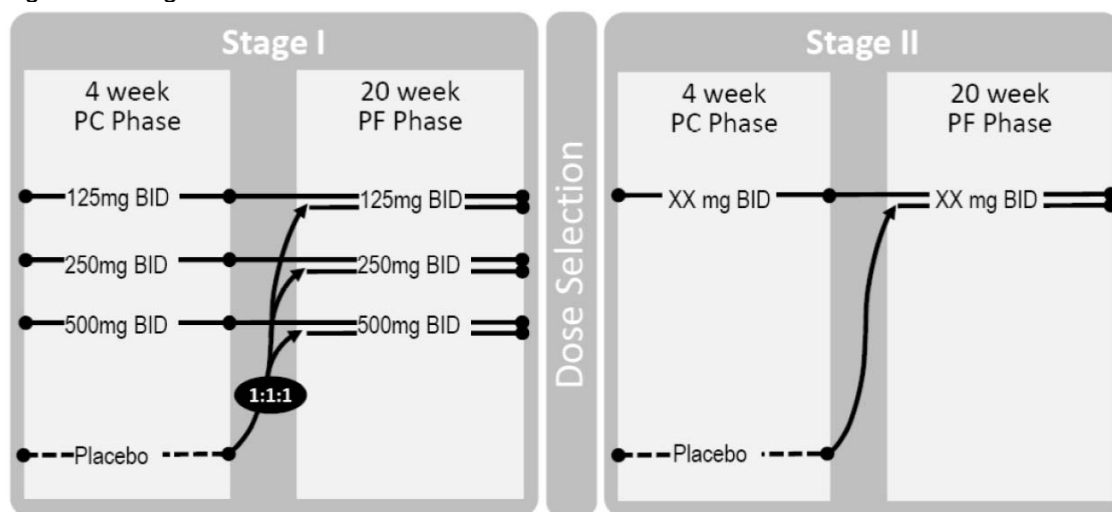
Given the low bioavailability and the inability to characterize crofelemer PK, a concentration-response relationship was not able to be determined. However, a dose-ranging phase was conducted as the first stage of the ADVENT trial. The doses that were selected for the dose-ranging portion of the study were 125 mg, 250 mg, and 500 mg twice daily. Patients were randomized to one of the three treatment regimens or to placebo for the first four weeks (PC phase). This was followed by a 20-week treatment period in which placebo-treated patients were re-randomized to one of the three active treatment groups. Patients who received crofelemer in the PC phase were maintained on that dose for the remaining 20-week period.

Figure 3. Overview of Stage I of the ADVENT trial



Following the PC phase, an interim analysis was conducted and the dose that appeared to show better efficacy and safety data was selected by an independent analysis committee for use in the second stage of the study. See Figure below.

Figure 4. Stages 1 and 2 of the ADVENT trial.



Abbreviation: PC = placebo-controlled; PF = placebo-free; and BID = twice daily.

The trial was composed of two stages, each made up of a 4-week PC phase and a 20-week PF phase. Multiple dose levels were compared only in Stage I and the lowest effective dose was selected for study in Stage II.

Table 7. Stage I clinical response

Parameter/Statistic ^{a, b}	Crofelemer 125 mg BID n (%)	Crofelemer 250 mg BID n (%)	Crofelemer 500 mg BID n (%)	Placebo BID n (%)
Responder – n/N (%)	9/44 (20.5%)	5/54 (9.3%)	9/46 (19.6%)	1/50 (2.0%)
Treatment Difference	18.5%	7.3%	17.6%	---
1-sided 97.5% CI for Diff.	[6.0%, ∞)	[-1.7%, ∞)	[5.3%, ∞)	---
1-sided p-value (vs. placebo)	0.0019	0.0563	0.0024	---

Abbreviations: BID = twice daily; CI = confidence interval; ITT = intent-to-treat.

Note: Clinical response was defined as ≤ 2 watery stools per week during at least 2 of the 4 weeks of the placebo-controlled treatment phase.

a P-values and CIs were calculated based on the methods of Posch and Bauer (2005).

b If less than 5 days of data were available in a week, the subject was classified as a non-responder for that week. Subjects who discontinued prematurely during the 4-week period were classified as non-responders. Subjects who used an ADM or opiate pain medication for > 3 days during the efficacy assessment period were also non-responders.

The results of the PC phase of the first stage of the study showed better clinical response in all treatment groups relative to placebo. However, there was no apparent dose response with the lowest dose group and the highest dose group showing similar response rates, while the middle dose group showed a lower response rate. Therefore, the lowest dose, 125 mg twice daily, was selected for the second stage of the trial and for marketing.

2.4.2 What are the characteristics of the exposure-response relationships (dose-response, concentration-response) for safety? If relevant, indicate the time to the onset and offset of the pharmacological response or clinical endpoint.

The only study in which multiple doses were compared head-to-head was the ADVENT trial. The table below shows safety data from the PC phase of Stage 1 of that trial.

Table 8. Summary of adverse events in the PC phase of Stage I of the ADVENT trial

Population	Crofelemer 125 mg BID n (%)	Crofelemer 250 mg BID n (%)	Crofelemer 500 mg BID n (%)	All Crofelemer n (%)	Placebo BID n (%)
Placebo-Controlled Treatment Phase					
N	130	54	42	226	137
Any TEAEs	45 (35)	10 (19)	7 (17)	62 (27)	45 (33)
Severe	2 (2)	1 (2)	0	3 (1)	5 (4)
Moderate	16 (12)	6 (11)	2 (5)	24 (11)	16 (12)
Mild	27 (21)	3 (6)	5 (12)	35 (16)	24 (18)
Drug-related TEAEs	9 (7)	2 (4)	2 (5)	13 (6)	5 (4)
TEAEs Leading to Discontinuation	0	0	0	0	4 (3)
SAEs	2 (2)	0	0	2 (<1)	4 (3)
Deaths	0	0	0	0	1 (<1)

Safety results show a slightly higher incidence of treatment emergent adverse events (TEAEs), severe TEAEs, TEAEs leading to discontinuation, and severe adverse events (SAEs) in the placebo group compared to the three crofelemer treatment groups. The incidence of mild or moderate TEAEs appears similar between the 125 mg crofelemer group and the placebo-treated group.

Table 9. Summary of adverse events in the PF phase of Stage I of the ADVENT trial

Population	Crofelemer 125 mg BID n (%)	Crofelemer 250 mg BID n (%)	Crofelemer 500 mg BID n (%)	All Crofelemer n (%)	Placebo BID n (%)
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Placebo-Free Treatment Phase				
N	220	67	50	337
Any TEAEs	130 (59)	34 (51)	27 (54)	191 (57)
Severe	11 (5)	2 (3)	2 (4)	15 (5)
Moderate	63 (29)	13 (19)	12 (24)	88 (26)
Mild	56 (26)	19 (28)	13 (26)	88 (26)
Drug-related TEAEs	21 (10)	3 (5)	4 (8)	28 (8)
TEAEs Leading to Discontinuation ^a	6 (3)	1 (2)	0	7 (2)
SAEs	9 (4)	3 (5)	1 (2)	13 (4)
Deaths	1 (<1)	0	0	1 (<1)

Source: Table 14.3.1.1 and 14.3.1.1pf, Section 14.3; corresponding Listing 16.2.7.1, Appendix 16.2.

Abbreviations: BID = twice daily; TEAE = treatment-emergent adverse event; and SAE = serious adverse event.

The incidence of mild, moderate, or severe TEAEs did not appear to correlate with the dose of crofelemer administered during this longer phase of the trial. In fact, the lowest dose group (125 mg twice daily) had a slightly higher incidence of SAEs, and TEAE, and drug-related TEAEs. However, the differences among these three groups were small.

Table 10. TEAEs Occurring in ≥ 3 and ≥ 5 Subjects Treated with Crofelemer or Placebo in the PC Phase and the PF Phase, respectively

System Organ Class Preferred Term	Crofelemer 125 mg BID n (%)	Crofelemer 250 mg BID n (%)	Crofelemer 500 mg BID n (%)	All Crofelemer n (%)	Placebo BID n (%)
Placebo- Controlled Treatment Phase					
N	130	54	42	226	137
Any TEAEs	45 (35)	10 (19)	7 (17)	62 (27)	45 (33)
Gastrointestinal Disorders					
Abdominal pain	2 (2)	0	1 (2)	3 (1)	1 (1)
Constipation	2 (2)	1 (2)	0	3 (1)	1 (1)
Dyspepsia	2 (2)	0	1 (2)	3 (1)	0
Flatulence	2 (2)	1 (2)	0	3 (1)	0
Nausea	1 (1)	1 (2)	1 (2)	3 (1)	1 (1)
General Disorders and Administration Site Conditions					
Fatigue	1 (1)	1 (2)	1 (2)	3 (1)	0
Infections and Infestations					
Upper respiratory tract infection	5 (4)	1 (2)	0	6 (3)	4 (3)
Urinary tract infection	3 (2)	1 (2)	0	4 (2)	1 (1)
Herpes zoster	2 (2)	1 (2)	0	3 (1)	0
Investigations					
Blood bilirubin increased	3 (2)	0	0	3 (1)	3 (2)
Blood bilirubin unconjugated increased	2 (2)	0	0	2 (1)	3 (2)
Musculoskeletal and Connective Tissue Disorders					
Arthralgia	2 (2)	0	1 (2)	3 (1)	0
Nervous System Disorders					
Headache	1 (1)	1 (2)	1 (2)	3 (1)	1 (1)
Placebo-Free Extension Phase					
N	220	67	50	337	
Any TEAEs	130 (59)	34 (51)	27 (54)	191 (57)	
Gastrointestinal Disorders					
Nausea	5 (2)	3 (5)	1 (2)	9 (3)	
Flatulence	6 (3)	0	1 (2)	7 (2)	
Infections and Infestations					
Upper respiratory tract infection	9 (4)	3 (5)	1 (2)	13 (4)	
Nasopharyngitis	5 (2)	3 (5)	3 (6)	11 (3)	
Bronchitis	8 (4)	0	2 (4)	10 (3)	
Gastroenteritis	4 (2)	2 (3)	0	6 (2)	
Sinusitis	4 (2)	1 (2)	1 (2)	6 (2)	
Urinary tract infection	3 (1)	1 (2)	1 (2)	5 (2)	

ALT increased	5 (2)	1 (2)	3 (6)	9 (3)
AST increased	3 (1)	1 (2)	4 (8)	8 (2)
Blood bilirubin increased	4 (2)	1 (2)	0	5 (2)
Musculoskeletal and Connective Tissue Disorders				
Back pain	5 (2)	1 (2)	1 (2)	7 (2)
Musculoskeletal pain	4 (2)	1 (2)	0	5 (2)
Nervous System Disorders				
Headache	3 (1)	1 (2)	2 (4)	6 (2)
Dizziness	3 (1)	0	2 (4)	5 (2)
Psychiatric Disorders				
Insomnia	3 (1)	1 (2)	1 (2)	5 (2)
Respiratory, Thoracic and Mediastinal Disorders				
Cough	8 (4)	3 (5)	0	11 (3)

Source: Table 14.3.1.2.1 and 14.3.1.2pf, Section 14.3; corresponding Listing 16.2.7.1, Appendix 16.2.

Abbreviations: ALT = alanine aminotransferase; AST = aspartate aminotransferase; BID = twice daily; PF = placebo-free; PC = placebo-controlled; and TEAE = treatment-emergent adverse event.

In the PC phase of Stage I, few AEs occur at a higher incidence in the crofelemer treated groups compared to the placebo group and the magnitude of the difference is very small. In the PF phase, in which there was no placebo group to compare, the AEs do not appear to be dose-related. Only the occurrence of nasopharyngitis and headache appear to be associated with increasing crofelemer dose in the PF phase.

2.4.3 Does this drug prolong the QT or QTc interval?

The effect of crofelemer on the QTc interval is not known. An in vitro study was conducted to assess the effect of crofelemer on the hERG K⁺ channel in transfected human embryonic kidney cells. The mean crofelemer IC₅₀ for the study was 135- to 1392-fold higher than the highest plasma concentration observed in the Phase 3 trials.

(b) (4)

2.4.4 Is the dose and dosing regimen consistent with the known relationship between dose-concentration-response, and are there any unresolved dosing or administration issues?

The proposed dose for marketing, 125 mg administered twice daily, is the lowest effective dose based on the results of the ADVENT trial. No dose-response or exposure-response relationship was able to be determined for this locally-acting drug.

2.5 Pharmacokinetics

2.5.1 What are the single dose and multiple dose PK parameters? (Provide tables to refer to in subsequent questions in this section)

The crofelemer pharmacokinetic parameters were not able to be well defined in healthy volunteers or HIV⁺ patients. Blood samples for crofelemer PK were drawn in nine Phase 1 studies and in the Phase 2 and Phase 3 trials. A mix of traditional (full) and sparse PK sampling was used for the Phase 1 studies while only sparse sampling was incorporated into the Phase 2 and Phase 3 trials. Given the low bioavailability of crofelemer, no model was able to be created to describe crofelemer PK parameters; therefore, the sparse sampling data collected during several studies has limited value. It should be noted that these studies were carried out over a long period of time (1992 – 2011) and the methods used in some of the earlier studies appear to have lacked the sensitivity to detect crofelemer. However, more sensitive analytical techniques, including LC/MS, were used in several later studies and were also unable to detect crofelemer in most samples.

Table 11. Summary of Crofelemer (SP-303) PK in Volunteers and HIV+ Patients

Study (Type)	PK sampling scheme	Maximum Daily Dose Studied	Proportion samples with SP-303	Max detectable SP-303	Detection Method (LOQ)
SP-303-I-01 (PK)	Full	10 mg/kg x 1	0/30 [*]	0	HPLC (0.5 µg/mL)
SP-303-I-02 (MTD)	Full	20 mg/kg x 1	0/8 [†]	0	HPLC (0.05 µg/mL)
SP-303-I-03 (MTD)	Full	600 mg QD x 7 days	Not Available	Not Available	HPLC
SP-303-I-04 (MTD)	Sparse	2000 mg QD x 7 days	0/17 [‡]	0	HPLC (0.05 µg/mL)
SP-303-I-07 (PK)	Full	1000 mg BID x 3.5 days	1/90	0.31	HPLC (0.3 µg/mL)
37554-103 (DDI)	Sparse	500 mg Q6H x 5 days	20/168	0.084	HPLC (0.05 µg/mL)
51818-101 (PK)	Full	1250 mg x 1	0/42	0	HPLC (0.4 µg/mL)
51818-102 (PK)	Sparse	500 mg Q6H x 2 days	0/192	0	HPLC (0.05 µg/mL)
CFFE1091 (Food Effect)	Full	500 mg x1	2/840	0.071	LC/MS/MS (0.05 µg/mL)
NP-303-101 (ADVENT)	Sparse	500 mg BID x 4 weeks	15/456 [#]	0.077	LC/MS/MS (0.05 µg/mL)
37554-209 (Phase 2)	None	500 mg Q6H x 4 days	No PK	--	--
37554-210 (Phase 3)	Sparse	500 mg Q6H x 27 days	11/504	0.8	HPLC (0.1 µg/mL)

* Only the blood samples from the highest dose group (10 mg/kg) were analyzed for SP-303

[†] Samples from only two subjects were analyzed for SP-303

[‡] Samples from only four subjects were analyzed for SP-303

[#] All blood samples from ADVENT were troughs

The results of the PK studies were difficult to interpret given the few samples with detectable crofelemer. This was further complicated by differences in the limit of detection across studies. However, even studies with the lowest LOQ of 50 ng/mL were found to have few detectable samples. Interestingly, blood samples collected in the first Phase 3 trial (37554-210, conducted in 1998), which had the highest LOQ (0.1 µg/mL), were found to have concentrations of crofelemer (up to 0.8 µg/mL) that were the highest of all the studies. Although a relatively high dose (500 mg QID) was used in that study, blood samples from other (Phase 1) studies using at least an equivalent daily dose were several times lower. A comparison between the PK results of trial 37554-210 and the ADVENT trial are complicated by the fact that only troughs were collected in the ADVENT trial. No PK was collected in the Phase 2 study in patients. Differences in analytical methodology across studies and the presence of proanthocyanidins in the diet may also account for the discrepancy in PK results.

2.5.2 How does the PK of the drug and its major active metabolites in healthy volunteers compare to that in patients?

The PK of crofelemer in healthy volunteers and patients was not able to be fully characterized due to the low apparent bioavailability; therefore differences in PK between the two populations could not be determined. Major active metabolites in humans have not been identified.

2.5.3 What are the characteristics of drug absorption?

Crofelemer has very low bioavailability. With the exception of monomers and dimers, proanthocyanidins are generally not absorbed [AJCN 2005;81:230s-42s].

Caco-2 cells were used to evaluate the potential for crofelemer to penetrate the small intestinal mucosa. The results demonstrated that crofelemer permeability was very low in the basolateral-to-apical and the apical-to-basolateral directions relative to the positive control.

Table 12. Permeability Coefficient of SP-3-3 (NP-3-3) in Caco-2 Cells

	A-B	B-A	B-A + Verapamil
Mannitol	0.6	0.8	ND
Caffeine	28.4	20.2	ND
Rhodamine	ND	7.0	0.0
NP 303 30 μ M	0.0	0.04	0.04
NP 303 100 μ M	0.0	0.02	0.02

ND = not determined

Results from the marker compounds including mannitol, caffeine, and rhodamine, confirmed the monolayers performed as expected. There was no change in the permeability of crofelemer in the presence of verapamil indicating crofelemer is not likely to be a P-gp substrate. The description of this study states that verapamil was added to the basolateral side of the cell monolayer. Ordinarily, verapamil would be added to the apical side of the monolayer as P-gp is present primarily in the apical membrane. However, verapamil is highly permeable and the results of the rhodamine portion of the study, indicating inhibition of P-gp when added to the basolateral side, provide some reassurance regarding the overall conclusions of the study.

2.5.4 What are the characteristics of drug distribution?

The drug distribution characteristics in vivo could not be determined. The in vitro free fraction of crofelemer in human plasma was determined using radiolabelled SP-303 with liquid scintillation counting of dialysate at eight hours. A preliminary experiment found that crofelemer has a low degree of non-specific binding to human plasma in vitro.

Table 13. Summary of SP-303 (NP-303) Plasma Protein Binding

[¹⁴ C] NP-303 Conc. (μ M)	Average % Binding	Average % Recovery
30	95.4	95.0
50	98.8	90.0

The results of this in vitro study indicate crofelemer is highly protein bound; however, the drug is poorly absorbed and the systemic exposure appears to be very low.

2.5.5 Does the mass balance study suggest renal or hepatic as the major route of elimination?

A mass balance study was not performed due to difficulty labeling the compound. The radiolabelled compound used in some of the in vitro studies was C¹⁴ *methylated* crofelemer. Therefore, the results observed in the in vitro studies may or may not be representative of the performance of unmethylated crofelemer.

2.5.6 What are the characteristics of drug metabolism?

Crofelemer metabolites were not characterized in clinical studies and no in vitro studies were conducted to assess the effect of various CYP isozymes on crofelemer. Proanthocyanidins are degraded into various aromatic acids by the gastrointestinal microflora. Some of these aromatic acids have been shown to increase in human urine following consumption of procyanidin-rich chocolate; however, microbial metabolism of a purified proanthocyanidin, such as crofelemer, has not been studied in humans.

Given the prevalence of CYP isozymes including 3A in the GI tract, it is reasonable to recommend a post-marketing commitment for the sponsor to study the effect of CYP enzymes on the metabolism of crofelemer in vitro.

An in vitro study was conducted to determine the effect of crofelemer on the metabolism of probe substrates of the CYP enzyme system (see 2.7.3.1).

2.5.7 What are the characteristics of drug excretion?

The excretion was not characterized.

2.5.8 Based on PK parameters, what is the degree of linearity or nonlinearity in the dose-concentration relationship?

These parameters were not able to be determined due to the low bioavailability of crofelemer.

2.5.9 How do the PK parameters change with time following chronic dosing?

Systemic crofelemer exposure is still low when administered for up to 24 weeks in patients; however, determination of precise PK parameters following single or multiple doses was not possible.

2.5.10 What is the inter- and intra-subject variability of PK parameters in volunteers and patients, and what are the major causes of variability?

These parameters were not able to be determined.

2.6 Intrinsic Factors

2.6.1 What intrinsic factors (age, gender, race, weight, height, disease, genetic polymorphism, pregnancy, and organ dysfunction) influence exposure and/or response and what is the impact of any differences in exposure on the pharmacodynamics?

The impact of intrinsic factors on crofelemer exposure or pharmacodynamics was not studied due to the difficulty characterizing PK in healthy volunteers and patients. The impact of race and gender were explored with regard to the primary efficacy endpoint in ADVENT. See 2.6.2.3 and 2.6.2.4.

- 2.6.2 Based upon what is known about exposure-response relationships and their variability, and the groups studied (volunteers vs. patients); what dosage regimen adjustments, if any, are recommended for each of these subgroups (examples shown below)? If dosage regimen adjustments are not based upon exposure-response relationships, describe the alternative basis for the recommendation.

No dosage adjustments are currently recommended. Bioavailability appears to be very low and no dose-response relationship has been established.

2.6.2.1 Elderly

No elderly subjects were included in the trials. Based on the low systemic exposure, no dosage adjustments for elderly patients is anticipated.

2.6.2.2 Pediatric patients. What is the status of pediatric studies and/or any pediatric plan for study?

The sponsor has submitted a request for deferral of pediatric studies for patients 0 - 18 years of age. They have also submitted a pediatric plan, (b) (4)



We recommend a rich sampling scheme, although the collection of blood samples may be staggered to reduce the overall number of samples and blood volume from each patient but be able to obtain data from more time points.

2.6.2.3 Gender

Fewer than 20% of patients in the ADVENT clinical trial were females and only 5-11% of the patients in the first clinical trial (37554-210) were females. Based on the results of the ADVENT trial, the proportion of responders receiving crofelemer (22/115) and placebo (9/116) in males favored crofelemer ($p = 0.0124$). For female patients however, there was no difference in the proportion of responders between crofelemer (2/22) and placebo (2/21) groups, likely due to the low numbers of these patients included in the ADVENT trial.

2.6.2.4 Race, in particular differences in exposure and/or response in Caucasians, African-Americans and/or Asians.

Approximately 16-28% of the participants in the ADVENT trial were African Americans. The remainder were largely classified as Caucasian or Hispanic (Hispanic was listed as an option for *race*, as opposed to an *ethnicity*, in the CRFs). The proportion of Caucasian responders was greater in the crofelemer (13/53) group compared to placebo (5/58). However, African American and Hispanic responders were grouped together in the study report. This combined group did not appear to respond as favorably to crofelemer (11/83) relative to placebo (6/80) when compared to the response in Caucasians. However, the overall numbers are low and no conclusions may be reached regarding racial differences in response to crofelemer.

2.6.2.5 Renal impairment

No studies were conducted in renally impaired patients.

2.6.2.6 Hepatic impairment

No studies were conducted in hepatically impaired patients.

2.6.2.7 What pregnancy and lactation use information is there in the application?

There are no studies of crofelemer in pregnant or nursing women. A rabbit study revealed no significant evidence of impaired fertility or harm to the fetus.

2.6.3 Does genetic variation impact exposure and/or response?

The impact of genetic polymorphisms on exposure and response were not evaluated.

2.7 Extrinsic Factors

2.7.1 What extrinsic factors (drugs, herbal products, diet, smoking, and alcohol use) influence exposure and/or response and what is the impact of any differences in exposure on pharmacodynamics?

The influence of these factors on clinical response or PD effect was not explored.

2.7.2 Based upon what is known about exposure-response relationships and their variability, what dosage regimen adjustments, if any, do you recommend for each of these factors?

No dosage adjustments are recommended based on extrinsic factors for this locally-acting drug.

2.7.3 Drug-Drug Interactions

2.7.3.1 Is there an in vitro basis to suspect in vivo drug-drug interactions?

Pooled human liver microsomes were used to investigate the effect of crofelemer on the metabolism of standard probes for CYP 1A2, 2B6, 2C9, 2C19, 2D6, 2E1, and 3A4 isozymes. Results from the in vitro study indicate that crofelemer has the potential to inhibit several CYP enzymes.

Table 14. Summary of Inhibition (*Non-Time Dependent*) of SP-303 (NP-303) on CYP Activity in Human Liver Microsomes

NP-303 Conc. (μM)	Percent (%) Inhibition Non-Time Dependent Samples							
	CYP1A2	CYP2B6	CYP2C9	CYP2C19	CYP2D6	CYP2E1	CYP3A4 (testosterone)	CYP3A4 (midazolam)
1	-16	-7	3	40	7	25	-7	3
10	35	29	20	53	51	33	33	20
50	87	79	94	92	100	53	94	94
75	87	83	94	92	100	53	94	94
Positive Control	87	74	94	92	98	39	90	94
Calculated IC ₅₀ (μM)	21.5	9.8	26.2	7.9	9.8	44.0	21.0	6.3

In vivo DDI studies are recommended based on the calculation of the theoretical GI inhibitor concentration [I_2] to in vitro inhibitory potential [IC_{50}]. A value exceeding 10 should prompt consideration of additional in vivo studies. The calculation is as follows:

$$\begin{aligned} \text{Dose} / 250\text{mL} &= \text{molar concentration in gut} \\ &= 240 \mu\text{M crofelemer (using average MW of 2100 Da)} \end{aligned}$$

240 μM crofelemer / IC_{50} (μM) < 10; further studies generally not recommended
 240 μM crofelemer / IC_{50} (μM) $\geq$ 10; further studies should be considered

The values calculated for crofelemer and the seven CYP enzymes in Table 14 were as follows: 11.1 (1A2), 24.5 (2B6), 9.1 (2C9), 30.3 (2C19), 24.4 (2D6), 5.5 (2E1), 11.4 (3A4 – testosterone), and 38.1 (3A4 – midazolam). Therefore, crofelemer may have the potential to inhibit 1A2, 2B6, 2C19, 2D6, and 3A4 at the concentration present in the gut. However, of these enzymes, only 3A4 is widely expressed in the small intestine, accounting for approximately 80% of total P450. Crofelemer does not appear to be an inhibitor of 1A2, 2C9, or 2E1. An in vivo DDI study with a 3A4 substrate is recommended.

Table 15. Summary of Inhibition (*Time Dependent*) of SP-303 (NP-303) on CYP Activity in Human Liver Microsomes

NP-303 Conc. (μM)	Percent (%) Inhibition Time Dependent Samples						
	CYP1A2	CYP2B6	CYP2C9	CYP2C19	CYP2D6	CYP2E1	CYP3A4 (testosterone)
10	33	22	-9	62	43	26	55
75	84	84	90	92	100	45	88
Calculated IC_{50} (μM)	31.7	39.4	48.7	8.1	18.0	83.3	9.1

The time-dependent inhibition for each isozyme was determined by pre-incubating crofelemer with the microsomes and NADPH for 30 minutes before initiating the reaction. An IC_{50} shift (> 2-fold lower) was only apparent for one of the two 3A4 substrates (testosterone) and not apparent for other CYP enzymes suggesting crofelemer does not exhibit time dependent inhibition. Interestingly, a shift to a higher IC_{50} value was observed for 2E1, 2D6, 2C9, 2B6, and 1A2. This may indicate that the concentration of crofelemer was too low in this study or that crofelemer is metabolically unstable.

Another in vitro study was conducted to characterize the effects of crofelemer on the induction of CYP enzymes (1A2, 2B6, and 3A) in cultured human hepatocytes. Three concentrations of crofelemer were used (10, 100, and 1000 nM) and compared to the effects of positive controls (omeprazole, phenobarbital, and rifampin) at appropriate concentrations.

Table 16. Summary of Induction Potential of SP-303 on CYPs 1A2, 2B6, and 3A in Human Hepatocytes

Treatment	Concentration	Enzyme activity (pmol/mg protein/min) ^a		
		Phenacetin O-dealkylation (CYP1A2)	Bupropion hydroxylation (CYP2B6)	Testosterone 6 β -hydroxylation (CYP3A4/5)
Dimethyl sulfoxide	0.1% (v/v)	29.0 $\pm$ 11.5	48.3 $\pm$ 45.8	2110 $\pm$ 1760
Crofelemer	10 nM	27.9 $\pm$ 10.0	45.2 $\pm$ 42.9	2130 $\pm$ 1790
Crofelemer	100 nM	28.6 $\pm$ 12.3	52.6 $\pm$ 54.0	1850 $\pm$ 1740
Crofelemer	1000 nM	29.1 $\pm$ 10.1	51.6 $\pm$ 49.9	2260 $\pm$ 1860
Omeprazole	100 nM	1100 $\pm$ 180	541 $\pm$ 769	4020 $\pm$ 1130
Phenobarbital	750 nM	56.0 $\pm$ 16.1	759 $\pm$ 1061	11800 $\pm$ 3400
Rifampin	10 nM	51.0 $\pm$ 24.5	353 $\pm$ 471	12100 $\pm$ 4100

a Values are the mean $\pm$ standard deviation of three determinations (human hepatocyte preparations H1001, H1005 and H1006).

Treatment of cultured human hepatocytes with crofelemer once daily for three days resulted in no increase in the activity of any of the three enzymes that were tested relative to the effects of the positive controls.

2.7.3.2 Are there other metabolic/transporter pathways that may be important?

Yes. The affinity of crofelemer for a variety of transporters was evaluated in vitro. A vesicular transport assay was performed with membrane vesicles prepared from SF9 ovarian cells and mammalian LLC-PK1 cells overexpressing transporters. Uptake transporter inhibition assays were performed using CHO, HEK293, and MDCKII cells expressing the transporters of interest. The results suggest that crofelemer has the potential to inhibit MRP2 (IC₅₀ 15 µM), MRP4 (IC₅₀ 7 µM), ASBT (IC₅₀ 7 µM), and OATP1A2 (IC₅₀ 7 µM) at concentrations expected in the gut. However, of these transporters, only MRP2 and OATP1A2 are expressed in enterocytes. Crofelemer is unlikely to inhibit OATP2B1 (IC₅₀ 39 µM) and PEPT1 (IC₅₀ 50 µM). Based on these results it appears that crofelemer has the potential to interact with drugs that are substrates of MRP2 and OATP1A2; however, these transporters have not been identified as mediating clinically significant drug-drug interactions and specific probes for these transporters have not been identified. Therefore, an in vivo drug interaction study is not recommended at this time.

2.7.3.3 Does the label specify co-administration of another drug (e.g., combination therapy in oncology) and, if so, has the interaction potential between these drugs been evaluated?

The label does not specify co-administration of crofelemer with other drugs; however, given the indication, it is likely to be administered with antiretroviral (ARV) drugs and with other antidiarrheal medications (ADMs). See 2.7.3.4 below.

2.7.3.4 What other co-medications are likely to be administered to the target patient population?

Crofelemer is likely to be administered with ARVs and with other ADMs. Given the mechanism of action and the lack of systemic absorption, the interaction with other ADMs is expected to be a pharmacodynamic interaction; crofelemer would be expected to increase the antidiarrheal effect of these coadministered drugs.

The potential for crofelemer to alter the exposure ARVs was explored in study 37554-103. This study included 28 healthy volunteers who received crofelemer 500 mg Q6H for five days. Three ARVs, nelfinavir, zidovudine, and lamivudine, were administered as a single dose together on Day 5. The results of the study showed crofelemer had no effect on the exposure of zidovudine and nelfinavir when administered as a single dose. However, there was a 20% decrease in lamivudine exposure when administered with crofelemer. A review of this study was also performed by Dr. Vikram Arya, who noted several limitations (see review dated 3 May 2012). His specific conclusions were as follows:

1. The dosing regimen of crofelemer used in the drug-drug interaction trial (500 mg every 6 hours) was different from the proposed dosing regimen (125 mg twice daily), hence, it is not clear if the effect of crofelemer 500 mg every 6 hour regimen on lamivudine pharmacokinetics will be similar to the effect of crofelemer 125 mg twice daily regimen (proposed regimen) on lamivudine pharmacokinetics.
2. There were differences in the composition of the formulation used in trial 37554-103 and pivotal clinical trial (ADVENT; NP303-101): Per the description provided in attachment 2 of the summary of pharmaceutical development (section 3.2.P.2; Page 30), the excipients in the core tablet used in trial NP303-101 included microcrystalline

cellulose, croscarmellose sodium, colloidal silicon dioxide and magnesium stearate. The excipients in the core tablet used in the drug-drug interaction trial included (b) (4). Crofelemer is not significantly absorbed into the systemic circulation; therefore, the potential effects of crofelemer on coadministered drugs in the gastrointestinal (GI) tract can vary depending on the active/inactive ingredients of the crofelemer formulation. In addition, the drug interaction trial included the use of crofelemer tablets contained within a capsule (b) (4), which is not the formulation intended for commercial use.

3. Use of single dose of the antiretroviral drugs does not represent the clinical scenario where multiple doses of these antiretroviral drugs are used.

The observed interaction with lamivudine is likely the result of an effect on absorption at the local level. Furthermore, the decrease in lamivudine mean systemic exposure was not considered by Dr. Arya to be clinically relevant and dosage adjustment was not recommended. Despite the limitations of the drug interaction study that were identified, additional drug-drug interaction studies with more commonly used ARVs were not recommended. According to Dr. Arya, the majority of drugs used for treatment of HIV-1 infection are either metabolized by CYP enzymes or eliminated unchanged by the kidney and because crofelemer is not significantly absorbed into the systemic circulation, it is not expected to alter the systemic concentrations of the antiretroviral drugs due to alteration of enzyme/transporter activity or compete for transporters in the renal proximal tubule.

2.7.3.5 Is there a known mechanistic basis for pharmacodynamic drug-drug interactions, if any?

Crofelemer may have an additive antidiarrheal effect when co-administered with other ADMs. Given the chronicity of secretory diarrhea in HIV⁺ patients taking ARVs and the negative impact on drug compliance, the benefit of this additive antidiarrheal effect likely outweighs the risks.

2.7.3.6 Are there any unresolved questions related to metabolism, active metabolites, metabolic drug interactions or protein binding?

The metabolic pathway of this polymer was not well characterized by the sponsor. However, the drug appears to have very low bioavailability and the activity is likely limited to the gastrointestinal tract.

2.7.4 What issues related to dose, dosing regimens or administration are unresolved, and represent significant omissions?

The timing of administration of ARVs relative to administration of crofelemer was not addressed. The sponsor concluded that the decrease in lamivudine exposure was not clinically relevant and; therefore, the drugs could be co-administered. Dr Arya also concluded that the decrease in lamivudine exposure was not clinically relevant but the timing of administration was not specifically addressed.

2.8 General Biopharmaceutics

2.8.1 What is the in vivo relationship of the proposed to-be-marketed formulation to the pivotal clinical trial formulation in terms of comparative exposure?

The crofelemer formulation used in the ADVENT trial is the same as the to-be-marketed formulation. Numerous other formulations were studied in the Phase 1, Phase 2, and

earlier clinical trial; however, no in vivo relationship between these formulations could be determined due to the low bioavailability.

2.8.2 What is the effect of food on the bioavailability (BA) of the drug from the dosage form? What dosing recommendation should be made, if any, regarding administration of the product in relation to meals or meal types?

Administration of crofelemer with a standard high-fat meal was not associated with an increase in systemic exposure of crofelemer in healthy volunteers. The sponsor performed a randomized, open-label, single-dose, 2-period, crossover study in healthy volunteers. Subjects were randomized at a 1:1 ratio to Group 1 (fasted then fed) or Group 2 (fed then fasted). Each subject received a single dose of crofelemer 500 mg (administered orally as 2 × 250 mg tablets) with a high fat meal and after fasting. Crofelemer was administered as a 500 mg (2 × 250 mg), enteric-coated tablet. This was the highest dose of crofelemer used in the clinical trials for HIV associated diarrhea and in studies conducted for other potential indications. The drug was administered with 240 mL of water in both groups but the fed group had to have completed their standard high-fat breakfast approximately 5 minutes prior to dosing. The tablet was manufactured by the same company and in the same manner as the 125 mg to-be-marketed tablet. The two study periods were separated by 7 days. All subjects fasted overnight prior to administration of crofelemer.

Blood samples for pharmacokinetic analyses of crofelemer were collected pre-dose and at 0.5, 1, 2, 4, 6, 8, 10, 12, 16, 24, 30, 36, 42, and 48 hours postdose following both fasted and fed single-dose treatment regimens. Two of 25 subjects had crofelemer plasma concentrations above the LOQ (50 ng/mL). Both detectable samples occurred in the fasting group; 54.7 ng/mL at 8 hours post dose in one subject and 70.7 ng/mL and 64.9 ng/mL at 4 and 6 hours post dose, respectively, in another subject. There was no detectable crofelemer in any plasma sample from subjects in the fed state.

These results indicate that there is no increase of crofelemer absorption when administered with a high fat meal. However, these results are difficult to interpret in the absence of an ability to accurately characterize PK, the lack of an exposure-response relationship, and the classification of this drug as one that is locally acting. In the ADVENT trial, crofelemer was administered one-half hour before the morning and evening meals, which may be reasonably described as taking the drug with a meal (as opposed to taking a drug in the fasted state, generally understood to be at least one hour before or two hours after a meal). Therefore, it seems reasonable to label the drug to be administered with or without a meal based on the design of the clinical trial and the supporting evidence from the food effect study.

2.8.3 Was the bioequivalence of the different strengths of the to be marketed formulation tested?

Only one strength of tablet, 125 mg, is currently planned for marketing.

2.8.4 What other significant, unresolved issues related to in vitro dissolution or in vivo BA and BE need to be addressed?

There are several issues related to in vitro dissolution testing. Please refer to ONDQA (Biopharm) review by Dr. Mark Seggel for a thorough description of these issues.

2.9 Analytical Section

2.9.1 How are the active moieties identified and measured in the plasma in the clinical pharmacology and biopharmaceutics studies?

Methods to detect crofelemer included HPLC and LC/MS. For this highly variable botanical drug, a number of methods were used throughout the clinical development program in an effort to maximize the extraction of the range of polymers expected while minimizing interference from the biological matrix. Studies NP-303-101 (ADVENT) and CFFE1091 employed a gel permeation chromatography (GPC) method with fluorescence detection to quantify plasma crofelemer.

2.9.2 Which metabolites have been selected for analysis and why?

One metabolite, 4'-O-methylepigallocatechin (4-MEGC), was selected for analysis in urine samples collected in studies 37554-103 and 37554-210 based on the identification of this metabolite in a study in rats. Crofelemer was not detected in quantifiable amounts (LOQ of 50 ng/mL) in the urine of any subject in study 37554-103 over the six hours post-dose. Due to irregularities in sample processing in study 37554-210, no urine data was reported. No other study included this metabolite in the analytical reports.

2.9.3 For all moieties measured, is free, bound or total measured? Is it appropriate?

Total crofelemer was measured in plasma and urine. This appears reasonable given the low bioavailability of the compound.

2.9.4 What bioanalytical methods are used to assess concentrations?

Methods to detect crofelemer included HPLC and LC/MS. See table below.

Table 17. Summary of Analytical Methods for SP-303 in Plasma

Study	Analytical Report (date)	Validated Method	Detection Method (LOQ)	LLOQ
SP-303-I-01	303-92-018 (1992)	TM-043	HPLC	0.5 µg/mL
SP-303-I-02	SP-303-G-007 (1993)	TM-018	HPLC	0.05 µg/mL
SP-303-I-03	SP-303-I-03 CSR (1993)	NA	HPLC	NA
SP-303-I-04	SP-303-G-007 (1993)	TM-018	HPLC	0.05 µg/mL
SP-303-I-07	SP-303-I-07 CSR (1995)	NA	HPLC	0.3 µg/mL
37554-103	048-98001D (1998)	048-98002V	HPLC	0.05 µg/mL
51818-101	NA	TM-043	HPLC	0.4 µg/mL
51818-102	SP-303-G-023 (1997)	TM-082	HPLC	0.05 µg/mL
CFFE1091	AA94739-01 (2010)	AA43544-01	LC/MS	0.05 µg/mL
NP-303-101 (ADVENT)	AA88803-01 (2011)	AA43544-01	LC/MS	0.05 µg/mL
37554-210	048-98002D (1999)	048-98003V	HPLC	0.1 µg/mL

2.9.4.1 What are the lower and upper limits of quantification (LLOQ/ ULOQ)?

What is the accuracy and precision at these limits?

The lower limit of quantitation was 50 ng/mL for SP-303 in plasma; however, at least four Phase 1 studies relied upon a method with a higher LOQ (100 to 500 ng/mL). The ULOQ ranged from 5000 to 10,000 ng/mL, depending on the study. There was wide variability in the analytical results with regard to precision and accuracy, which illustrates the challenges that existed with the methodology in the early studies. In the two studies that used mass spectrometry (ADVENT and CFFE1091), the between-batch accuracy ranged from (b) (4) and the precision was (b) (4). For calibration standard concentrations, accuracy ranged from (b) (4) and precision was (b) (4). The majority of the analytical reports from studies relying upon HPLC did not contain sufficient information to describe these parameters, including the study that reported 4-MEGC analysis from urine.

Given that crofelemer is a polymeric compound with linear chains of 1-28 units, it is reasonable to expect that there would be a distribution of molecular weights (corresponding to polymer length) that would be reproducible when analyzed by a chosen analytical method. Unfortunately, despite use of a variety of analytical methods to quantify crofelemer in plasma, no method was able to separate and quantify particular polymers. The molecular weight range for the GPC column was 100-10,000 Daltons, which was designed to allow the passage of the complete range of polymers present in the reference material. However, due to a variety of issues including the presence of numerous -OH groups causing prolonged retention of the sample, the potential interference due to ingestion of proanthocyanidin monomers as part of a subjects' normal diet, the multiple charges associated with crofelemer, and the inherent product content variability, the preferred analytical method resulted in the elution of one common peak.

The absence of complete information from some analytical reports raises questions about the interpretation of the PK data, particularly the earlier studies using HPLC. However, data from the literature supports the conclusion that systemic exposure to orally administered proanthocyanidins is very low. This does not imply that some additional effort to characterize the systemic exposure is not necessary or would be feasible in the event future studies are recommended.

2.9.4.2 What is the sample stability under the conditions used in the study? (long-term, freeze-thaw, sample-handling, sample transport, autosampler)

Crofelemer is stable for up to (b) (4) days when stored in (b) (4) at (b) (4); however, some plasma samples were frozen for more than 4 years prior to analysis in the ADVENT trial. To address this issue, the sponsor is performing an extended long-term stability evaluation. Results from the ADVENT trial may be impacted by the long-term stability evaluation; however, the results of the food effect study (CFFE1091) are not affected.

3 Detailed Labeling Recommendations

Section 12 CLINICAL PHARMACOLOGY

Sponsor's proposed label

12.1 Mechanism of Action

(b) (4)

(b) (4)



Reviewer's proposed label

12.1 Mechanism of Action

(b) (4)



12.3 Pharmacokinetics

(b) (4)



4 Appendices

4.1 Consult Review (including Pharmacometric Reviews)

See attached DAARTS document by Dr. Vikram Arya dated 3 May 2012 for a complete review of the drug-drug interaction study (37554-103).

To	Kristina E. Estes, Pharm.D. Division of Clinical Pharmacology 3 Office of Clinical Pharmacology
From	Vikram Arya, Ph.D., Clinical Pharmacology Reviewer Division of Clinical Pharmacology 4 Office of Clinical Pharmacology Shirley Seo, Acting Clinical Pharmacology Team Leader, Division of Clinical Pharmacology 4 Office of Clinical Pharmacology
Name of drug product (NDA #)	Crofelemer (NDA 202292)
Sponsor	Salix Pharmaceuticals Raleigh, North Carolina
Re:	Drug interactions with Antiretroviral Drugs (Lamivudine, Zidovudine, Nelfinavir)
Date of consult request	Feb 21, 2012

Background:

This document is a consult review of a drug interaction study conducted with crofelemer and three antiretroviral drugs (lamivudine, zidovudine, nelfinavir) as requested by the gastrointestinal products team within the Office of Clinical Pharmacology.

Crofelemer is an inhibitor of both the cyclic adenosine monophosphate (cAMP)-stimulated cystic fibrosis transmembrane conductance regulator (CFTR) chloride ion (Cl⁻) channel and calcium activated Cl⁻ channels (CaCC) at the luminal membrane of the enterocytes. CFTR Cl⁻ and CaCC regulate Cl⁻ and fluid secretion by intestinal epithelial cells.

The proposed indication of crofelemer is control and symptomatic relief of diarrhea in patients with HIV/AIDS who are taking antiretroviral therapy. The proposed dose is 125 mg twice a day, with or without food using 125 mg tablets.

Summary of Drug-Drug Interaction Trial (Trial 37554-103):

A drug-drug interaction trial was conducted to evaluate the effect of multiple doses of SP-303 (crofelemer) on the pharmacokinetics of the antiviral drugs zidovudine, lamivudine, and nelfinavir in healthy subjects (trial 37554-103).

Trial 37554-103 was a randomized, double-blind, placebo-controlled, cross-over, pharmacokinetic trial. The primary objective of the trial was to determine whether crofelemer, administered orally (500 mg every 6 hours) over 5 consecutive 24-

hour periods (20 doses) affects the pharmacokinetic parameters of zidovudine, lamivudine, and nelfinavir.

Following screening, there were two in-clinic 7-day treatment/surveillance periods with a 7-day washout period between the two periods. A final evaluation was conducted 4 days after the end of the second treatment/surveillance period. The total duration of the trial was 25 days.

28 subjects were randomized such that 14 subjects received crofelemer and 14 subjects received placebo in the first treatment/surveillance period; each group received the alternate regimen in the second treatment/surveillance period. During each treatment/surveillance period, the subjects received crofelemer (one 500 mg capsule every 6 hours; delayed release tablets in capsules was used in the trial) or placebo (one capsule; tablets in capsule was used in the trial) every 6 hours until they received 20 doses. On the morning of the last day of treatment, shortly after the 19th dose of crofelemer was administered, the subjects received a single dose of zidovudine (formulation not specified), lamivudine (tablet, 1 X 150 mg), and nelfinavir (tablets, 3 X 250 mg). Zidovudine and lamivudine were administered before breakfast (breakfast was served 1 hour and 40 minutes later) and nelfinavir was administered 20 minutes after breakfast.

Serial blood samples (relative to the 19th dose of crofelemer) were collected to determine the plasma concentrations of crofelemer (samples collected at pre-dose, 3 and 6 hours post dose) and each antiviral drug (zidovudine: samples collected at pre-dose and up to 9 hours post dose; lamivudine: samples collected at pre-dose and up to 25 hours post dose; nelfinavir: samples collected at pre-dose and up to 23 hours post dose)

Table 1 shows the mean pharmacokinetic parameters of the antiviral agents.

Table 1: Mean pharmacokinetic parameters of zidovudine, lamivudine, and Nelfinavir after co-administration of multiple dose crofelemer and single dose of zidovudine, lamivudine, and nelfinavir

Mean ($\pm$ SD) Pharmacokinetic Parameters for Antiretroviral Agents

<u>Antiretroviral Agent</u>	<u>Treatment</u>	<u>AUC_{0-∞}</u> ($\mu\text{g}\cdot\text{h}/\text{mL}$)	<u>C_{max}</u> ($\mu\text{g}/\text{mL}$)	<u>T_{max}</u> (h)	<u>t_{1/2}</u> (h)
<u>Zidovudine</u>	<u>Placebo</u>	<u>2.62 (0.46)</u>	<u>1.79 (0.67)</u>	<u>0.64 (0.20)</u>	<u>1.13 (0.24)</u>
	<u>Crofelemer</u>	<u>2.69 (0.53)</u>	<u>1.75 (0.55)</u>	<u>0.86 (0.51)^b</u>	<u>1.13 (0.17)</u>
	<u>95% CI^a</u>	<u>97-108%</u>	<u>85-114%</u>	<u>103-153%</u>	<u>91-110%</u>
<u>Lamivudine</u>	<u>Placebo</u>	<u>6.19 (1.49)</u>	<u>1.51 (0.61)</u>	<u>0.98 (0.52)</u>	<u>6.73 (2.12)</u>
	<u>Crofelemer</u>	<u>4.89 (1.62)^b</u>	<u>1.23 (0.49)^b</u>	<u>0.90 (0.42)</u>	<u>7.03 (2.03)</u>
	<u>95% CI^a</u>	<u>64-90%</u>	<u>68-93%</u>	<u>75-114%</u>	<u>91-119%</u>
<u>Nelfinavir</u>	<u>Placebo</u>	<u>32.1 (12.5)</u>	<u>3.45 (0.99)</u>	<u>4.57 (0.92)</u>	<u>3.65 (0.88)</u>
	<u>Crofelemer</u>	<u>33.5 (10.9)</u>	<u>3.36 (0.77)</u>	<u>4.86 (1.35)</u>	<u>3.61 (1.02)</u>
	<u>95% CI^a</u>	<u>99-113%</u>	<u>92-105%</u>	<u>97-113%</u>	<u>93-103%</u>

Abbreviations: AUC_{0-∞} = area under the curve from time 0 to infinity; CI = confidence interval; C_{max} = maximum observed plasma concentration; SD = standard deviation; T_{max} = time to maximum observed plasma concentration; t_{1/2} = terminal half-life.

a Confidence interval expressed as percent of placebo

b Statistically significant difference from placebo

The mean $AUC_{0-\infty}$ of lamivudine decreased by ~21 % when a single dose of lamivudine was co-administered with multiple doses of crofelemer as compared with when a single dose of lamivudine was administered with placebo. The mean $AUC_{0-\infty}$ values for zidovudine and nelfinavir were not significantly altered when a single dose of zidovudine and nelfinavir was co-administered with multiple doses of crofelemer as compared with a single dose of zidovudine and nelfinavir administered with placebo.

Summary of Crofelemer Pharmacokinetics (Per the package insert submitted with the NDA application)

-  (b) (4)

Response to Consult Questions

Consult Question # 1:

Overall study design - how well was the study designed to address questions about ARV therapy in general? Were the selected drugs appropriate? If the study is not acceptable (or interpretable, etc.) would you recommend any additional studies as a PMR?

Response:

Trial Design:

Overall, the trial design has several limitations in its interpretation such as:

- The dosing regimen of Crofelemer used in the drug-drug interaction trial 37554-103 (500 mg every 6 hours) is different from the proposed dosing regimen (125 mg twice daily), hence, it is not clear if the effect of crofelemer 500 mg every 6 hour regimen on lamivudine pharmacokinetics will be similar to the effect of crofelemer 125 mg twice daily regimen (proposed regimen) on lamivudine pharmacokinetics.
- Differences in the composition of the formulation used in trial 37554-103 and pivotal clinical trial (ADVENT; NP303-101): Per the description provided in attachment 2 of the summary of pharmaceutical development (section 3.2.P.2; Page 30), the excipients in the core tablet used in trial

NP303-101 included microcrystalline cellulose, croscarmellose sodium, colloidal silicon dioxide and magnesium stearate. The excipients in the core tablet used in the drug-drug interaction trial included (b) (4)

Crofelemer is not significantly absorbed into the systemic circulation; therefore, the potential effects of crofelemer on co-administered drugs in the gastrointestinal (GI) tract can vary depending on the active/inactive ingredients of the crofelemer formulation. In addition, the drug interaction trial included the use of crofelemer tablets contained within a capsule (b) (4)

, which is not the formulation intended for commercial use.

- Use of single dose of the antiretroviral drugs does not represent the clinical scenario where multiple doses of these antiretroviral drugs are used.

Selection of Antiretroviral Drugs:

The drug-drug interaction trial (37554-103) evaluated the effect of multiple dose crofelemer on single-dose pharmacokinetics of lamivudine (nucleoside reverse transcriptase inhibitor), zidovudine (nucleoside reverse transcriptase inhibitor), and nelfinavir (protease inhibitor). The trial was conducted in May-June 1998. Since that time, several new HIV drugs with novel mechanisms of action have been approved. Hence, antiretroviral drugs used in trial 37554-103 do not represent the preferred antiretroviral drugs used in current clinical practice. For example, in the guidelines for the Use of Antiretroviral Agents in HIV-1 Infected Adults and Adolescents (April 2012 version) developed by the Department of Health and Human Services (DHHS), the majority of “preferred regimens” do not recommend the use of any of the antiretroviral agents used in the drug-drug interaction trial. Further, neither the “preferred” nor the “alternative” regimens recommends the use of nelfinavir.

Need for Additional Trials:

The majority of drugs used for treatment of HIV-1 infection are either metabolized by CYP enzymes or eliminated unchanged by the kidney. As crofelemer is not significantly absorbed into the systemic circulation, crofelemer is not expected to alter the systemic concentrations of the antiretroviral drugs due to alteration of enzyme/transporter activity or compete for transporters in the renal proximal tubule. Thus, the observed interaction with lamivudine is likely the result of an effect on absorption at the local level.

No conclusions can be drawn regarding the potential effect of crofelemer on systemic exposures of other antiretroviral drugs from trial 37554-103. Extrapolating the findings of trial 37554-103 to other “crofelemer-antiretroviral drug” combinations is challenging because the potential for a local interaction between crofelemer and other antiretroviral drugs can vary from one drug to another.

Consult Question # 2:

Regarding the lamivudine exposure - is the 20% reduction in lamivudine exposure potentially clinically significant? Of so, do you have any recommendations, through labeling or PMRs, to address the concern?

A 20% decrease in the mean systemic exposures of lamivudine is not considered to be clinically relevant. In support of this conclusion, per the approved package insert of Ziagen (Abacavir), co-administration of abacavir and lamivudine resulted in a mean decrease in lamivudine exposure (~15 %) similar to the decrease in the mean exposure of lamivudine in the drug-drug interaction trial with crofelemer. The abacavir package insert indicates that a decrease in the mean lamivudine exposure is not considered clinically relevant.

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/s/

VIKRAM ARYA
05/02/2012

SHIRLEY K SEO
05/03/2012

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/s/

KRISTINA E ESTES
08/31/2012

SUE CHIH H LEE
08/31/2012

EDWARD D BASHAW
09/04/2012

BIOPHARMACEUTICS REVIEW Office of New Drug Quality Assessment			
Application No.:	NDA 202-292	Reviewer: Mark R. Seggel	
Submission Date:	05-DEC-2011		
Division:	DGIEP	Team Leader: Angelica Dorantes, Ph.D.	
Applicant:	Salix Pharmaceuticals, Inc.		
Trade Name:	<i>To be determined</i>	Date Assigned:	06-JAN-2012
Generic Name:	Crofelemer	Date of Review:	31-JUL-2012
Indication:	Control and symptomatic relief of secretory diarrhea in HIV/AIDS patients on anti-retroviral therapy	Type of Submission:	505(b)(1); NME; priority review (with 3-month extension)
Formulation / strengths	125 mg Delayed-Release Tablet	GRMP Goal:	08-AUG-2012
Route of Administration	Oral; 125 mg <i>bid</i>	PDUFA Goal:	05-SEP-2012

SUMMARY:

Submission: NDA 202-292 for Crofelemer Delayed-Release Tablets was submitted on 06-DEC-2011 as a 505(b)(1) submission. Crofelemer is a new molecular entity (NME). It is a botanical extract composed of proanthrocyanidin oligomers with an average molecular weight range of 1700 - 2500 Daltons. Oligomers consist of from 3 to 30 monomers, with an average chain length of approximately 7 units. Crofelemer is highly soluble (18 - 20 mg/mL) in water over a range of pH (1.1 - 7.4).

Crofelemer is indicated for the treatment of secretory diarrhea in HIV-infected patients. It is thought to work locally by blocking chloride ion secretion and associated water loss in the small intestine. However, the structural feature(s) of the oligomers that contribute to its activity are currently unknown. Absorption of oligomers of three or more units from the GI tract reportedly is minimal. There are negligible toxicities associated with crofelemer.

Review: The Biopharmaceutics review is focused on the evaluation and approvability of the proposed dissolution method and acceptance criteria.

Biopharmaceutics Information:

Formulation: The proposed commercial drug product consists of a tablet core containing 125 mg of crofelemer, microcrystalline cellulose, croscarmellose sodium, silicon dioxide and magnesium stearate. The tablet core is coated (b) (4)

Dissolution Test and Acceptance Criteria: Initially Salix proposed (b) (4)

(b) (4) Salix was advised to evaluate the product's performance (*drug release*) in accordance with the standard dissolution testing conditions described in "USP<711> **Dissolution, Method A for Delayed-Release Dosage Forms**". Each test consists of an acid stage and a buffer stage. While Salix agreed to the USP test, they proposed for the acid stage a limit of NMT (b) (4) after a (b) (4) hour, and a limit of $Q=7^{(b)(4)}$ at (b) (4) minutes in the pH 6.8 buffer stage. While the available dissolution data are admittedly limited, it appears that limits of NMT (b) (4) dissolved at 2 hours and a $Q = (b) (4)$ at 45 minutes, are achievable for the acid and buffer stages, respectively.

RECOMMENDATION:

Based on the evaluation of the overall dissolution information/data, the following dissolution testing conditions and acceptance criteria have been recommend for Crofelemer-Delayed Release Tablets.

Crofelemer 125 mg Delayed-Release Tablets Dissolution Method and Acceptance Criteria					
Apparatus	Rotation Speed	Medium Volume	Temperature	Medium	Acceptance criteria
USP II (paddle)	75 rpm	750 ml	37°C	<u>Acid Stage 1*:</u> 0.1 N HCl	NMT (b) (4) at 2 hrs
USP II (paddle)	75 rpm	1000 ml	37°C	<u>Buffer Stage 1*:</u> Sodium phosphate buffer, pH 6.8	$Q = (b) (4)$ at 45 minutes

*Assay by UV-spectrophotometry at 280 nm

Note that the above recommendation was conveyed to the Applicant in an IR letter dated 20-JUL-2012. The Applicant agreed verbally during 26-JUL-2012 teleconference (written response submitted 31-JUL-2012) to the above dissolution regulatory test and acceptance criteria. In this teleconference the Applicant also agreed to continue to collect dissolution profiles by both, the original dissolution methodology and by the new test. The dissolution profiles will be submitted in a Prior Approval Supplement (PAS) to the NDA providing for deletion of the original test.

(b) (4)
they have agreed to perform the testing as per USP and also they agreed to implement the more meaningful acceptance criteria listed in the above table.

It is therefore recommended from the Biopharmaceutics perspective, that NDA 202-292 for Crofelemer Delayed-Release Tablets be APPROVED as amended.

Signature

Mark R. Seggel
Biopharmaceutics Reviewer
Office of New Drug Quality Assessment

Signature

Angelica Dorantes, Ph.D.
Biopharmaceutics Team Leader
Office of New Drug Quality Assessment

BIOPHARMACEUTICS ASSESMENT

REVIEW NOTES

Documents Reviewed

Submission Date	eCTD Sequence
05-DEC-2011	0000
24-FEB-2012	0004
06-APR-2012	0009
30-APR-2012	0011
01-JUN-2012	0013
29-JUN-2012	0015
31-JUL-2012	0020

Relevant Communications

Type	Date
Information Request	07-FEB-2012
Filing Communication	17-FEB-2012
Teleconference	15-MAR-2012
Information Request	20-MAR-2012
Teleconference	06-APR-2012
Information Request / Advice	26-APR-2012
Information Request	20-JUL-2012
Teleconference	26-JUL-2012

Drug Product Components and Composition

Table 1: Components and Composition of Crofelemer Tablets, 125 mg

Ingredient	Reference to Standard	Function	Theoretical Quantity (mg/unit dose)
Uncoated Tablet			
Crofelemer	In-House	Active	125.00 ^a
Microcrystalline Cellulose (b) (4)	NF		(b) (4)
Croscarmellose Sodium	NF		
Colloidal Silicon Dioxide	NF		
Magnesium Stearate	NF		
Uncoated Tablet Weight:		---	
Coating^c			
Ethyl Acrylate and Methacrylate Copolymer Dispersion (b) (4)	NF		(b) (4)
White Dispersion (b) (4)	Supplier		
Talc (b) (4)	USP		
Triethyl Citrate	NF		
(b) (4)	USP		
Total Theoretical Weight:		---	550.0
(b) (4)			

Reviewer Comments:

(b) (4)
[REDACTED]
[REDACTED] *Salix decided to imprint the tablet with ink. This should not affect tablet performance.*

Pharmaceutical Development:

Background: [REDACTED] (b) (4)
[REDACTED]

IND 51,818 was submitted in November 1996 for the treatment [REDACTED] (b) (4). In April 2003 the IND was transferred to Napo Pharmaceuticals (previously dba PS Pharmaceuticals). In December 2009, crofelemer was licensed to Salix Pharmaceuticals; IND 51,818 was transferred as well. Virtually all pharmaceutical development appears to have been completed by Napo. {The status of the license agreement is the subject of current litigation in NY State Supreme Court.}

The earliest formulations studied were oral solutions containing crofelemer. Subsequently, encapsulated enteric-coated beads were developed. Ultimately, enteric-coated tablets containing 125-, 250- and 500-mg crofelemer were prepared. All enteric-coated formulations appear to have used [REDACTED] (b) (4).

Enteric Coating Formulation Development:

The first coating formulation used to coat crofelemer beads was comprised of [REDACTED] (b) (4). The coating used for the initial 250- and 500-mg tablet formulations was comprised of [REDACTED] (b) (4). [REDACTED] Patheon, the contract drug product manufacturer, conducted a series of coating studies to determine the optimum coating conditions and amount of coating needed to achieve suitable acid resistance and dissolution results. It was concluded that an overage weight gain of [REDACTED] (b) (4) enteric-coating provided adequate acid resistance and subsequent dissolution.

[REDACTED] (b) (4)

Salix concludes, [REDACTED] (b) (4)

(P.2, p. 20).

Elsewhere (P.5.6.8, Justification of Specification - Dissolution), Salix refers to [REDACTED] (b) (4)

Reviewer Comments:

- 1 *In U.S. Patent 7,341,744 B1, issued March 11, 2008 to Napo Pharmaceuticals, it is reported that after 0.03 hours in HCl, the peak area of the proanthocyanidin polymer composition profile, i.e. the concentration of the proanthocyanidin polymer composition, was reduced by 19%. After 0.5 hours and 2.0 hours incubation with HCl, the peak area of the proanthocyanidin polymer profile was reduced by 28% and 26%, respectively. These results indicate that most of the decrease of the proanthocyanidin polymer composition due to HCl exposure occurred within the first 2-3 minutes of incubation. The binding of the Proanthocyanidin Polymer Composition (i.e., crofelemer) with pepsin in simulated gastric fluid is also reported in the patent.*
- 2 *Recently, Dr. S. Chakder, Supervisory Pharmacologist, DGIEP, remarked that the interaction of crofelemer with stomach acid and pepsin were previously highlighted in the meeting minutes of a teleconference with Napo regarding IND 51,818. In addition, the precipitation of crofelemer in the presence of pepsin was documented in Shaman Pharmaceuticals Report No. SP-303-A-059 (dated July 1998). Salix included both documents in their NDA submission.*
- 3 *To further address the inconsistency regarding the stability of crofelemer in the acidic conditions of the stomach, Salix was asked to re-evaluate this issue (see 26-APR-2012 Information Request). The stability of crofelemer under acidic conditions was re-assessed using three separate analytical procedures (HPLC assay, related substances by HPLC, oligomer composition procedure). The results submitted June 29, 2012, (eCTD 0015) suggest that crofelemer remains stable under acidic conditions. Of course this depends on the adequacy of the tests to detect meaningful changes.*
- 4 *Whether any of the potential effects of stomach acid or pepsin on crofelemer would have any impact on the safety and efficacy of the locally acting drug product is unknown. Nevertheless, [REDACTED] (b) (4) the requisite for a consistently functioning enteric-coating on the tablet appears to be justified. In the absence of other information, adequate quality control of the enteric-coating is justified. Product labeling should reflect that the drug product is indeed a delayed-release formulation, and should include typical precautions associated with delayed-release products (e.g., do not crush or chew). No alcohol dose-dumping data were submitted; although this unlikely to result in adverse events, the labeling should warn against taking the product with alcohol.*

Dissolution Method Development: (eCTD sections 2.7.1, 3.2.P.2, asrpt-112, and asrpt-132)

(b) (4)

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(b) (4)

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/s/

MARK R SEGCEL
08/01/2012

ANGELICA DORANTES
08/01/2012

CLINICAL PHARMACOLOGY AND BIOPHARMACEUTICS FILING FORM/CHECKLIST FOR NDA/BLA or Supplement

Office of Clinical Pharmacology

New Drug Application Filing and Review Form

General Information About the Submission

	Information		Information
NDA/BLA Number	2022421	Brand Name	unknown
OCP Division (I, II, III, IV, V)	III	Generic Name	Crofelemer
Medical Division	DGIEP	Drug Class	Anti-diarrheal
OCP Reviewer	Kris Estes	Indication(s)	Diarrhea in patients with HIV on ARVs
OCP Team Leader	Sue Chih Lee	Dosage Form	Tablet
Pharmacometrics Reviewer		Dosing Regimen	125 mg BID
Date of Submission	6 December 2011	Route of Administration	PO
Estimated Due Date of OCP Review	14 April 2012	Sponsor	Salix
Medical Division Due Date		Priority Classification	Priority
PDUFA Due Date	5 June 2012		

Clin. Pharm. and Biopharm. Information

	"X" if included at filing	Number of studies submitted	Number of studies reviewed	Critical Comments If any
STUDY TYPE				
Table of Contents present and sufficient to locate reports, tables, data, etc.	X			
Tabular Listing of All Human Studies	X			
HPK Summary	X			
Labeling	X			
Reference Bioanalytical and Analytical Methods	X			
I. Clinical Pharmacology				
Mass balance:				
Isozyme characterization:				
Blood/plasma ratio:				
Plasma protein binding:	X	1		
Pharmacokinetics (e.g., Phase I) -				
Healthy Volunteers-				
single dose:	X	3		
multiple dose:	X	2		
Patients-				
single dose:				
multiple dose:	X	2 (as part of Phase 3 trials)		
Dose proportionality -				
fasting / non-fasting single dose:				
fasting / non-fasting multiple dose:				
Drug-drug interaction studies -				
In-vivo effects on primary drug:				
In-vivo effects of primary drug:	X	1		
In-vitro:	X	1		
Subpopulation studies -				
ethnicity:				
gender:				
pediatrics:				

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CLINICAL PHARMACOLOGY AND BIOPHARMACEUTICS FILING FORM/CHECKLIST FOR NDA/BLA or Supplement

geriatrics:				
renal impairment:				
hepatic impairment:				
PD -				
Phase 2:				
Phase 3:				
PK/PD -				
Phase 1 and/or 2, proof of concept:				
Phase 3 clinical trial:	X	10 (3 for HIV pts)		
Population Analyses -				
Data rich:				
Data sparse:				
II. Biopharmaceutics				
Absolute bioavailability				
Relative bioavailability -				
solution as reference:				
alternate formulation as reference:				
Bioequivalence studies -				
traditional design; single / multi dose:				
replicate design; single / multi dose:				
Food-drug interaction studies	X	1		
Bio-waiver request based on BCS				
BCS class				
Dissolution study to evaluate alcohol induced dose-dumping				
III. Other CPB Studies				
Genotype/phenotype studies				
Chronopharmacokinetics				
Pediatric development plan	X	1		
Literature References				
Total Number of Studies		13		

On **initial** review of the NDA/BLA application for filing:

	Content Parameter	Yes	No	N/A	Comment
Criteria for Refusal to File (RTF)					
1	Has the applicant submitted bioequivalence data comparing to-be-marketed product(s) and those used in the pivotal clinical trials?			X	TBM product used in pivotal Phase 3 trial
2	Has the applicant provided metabolism and drug-drug interaction information?	X			
3	Has the sponsor submitted bioavailability data satisfying the CFR requirements?	X			
4	Did the sponsor submit data to allow the evaluation of the validity of the analytical assay?	X			
5	Has a rationale for dose selection been submitted?	X			
6	Is the clinical pharmacology and biopharmaceutics section of the NDA organized, indexed and paginated in a manner to allow substantive review to begin?	X			
7	Is the clinical pharmacology and biopharmaceutics section of the NDA legible so that a substantive review can begin?	X			
8	Is the electronic submission searchable, does it have appropriate hyperlinks and do the hyperlinks work?	X			

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Criteria for Assessing Quality of an NDA (Preliminary Assessment of Quality)					
Data					
9	Are the data sets, as requested during pre-submission discussions, submitted in the appropriate format (e.g., CDISC)?	X			
10	If applicable, are the pharmacogenomic data sets submitted in the appropriate format?			X	
Studies and Analyses					
11	Is the appropriate pharmacokinetic information submitted?		X		No mass balance study - not able to radiolabel product. (b) (4)
12	Has the applicant made an appropriate attempt to determine reasonable dose individualization strategies for this product (i.e., appropriately designed and analyzed dose-ranging or pivotal studies)?	X			
13	Are the appropriate exposure-response (for desired and undesired effects) analyses conducted and submitted as described in the Exposure-Response guidance?		X		Very low BA, E-R relationship not able to be determined
14	Is there an adequate attempt by the applicant to use exposure-response relationships in order to assess the need for dose adjustments for intrinsic/extrinsic factors that might affect the pharmacokinetic or pharmacodynamics?		X		Not possible to base dose adjustments on exposure given low BA
15	Are the pediatric exclusivity studies adequately designed to demonstrate effectiveness, if the drug is indeed effective?			X	
16	Did the applicant submit all the pediatric exclusivity data, as described in the WR?			X	
17	Is there adequate information on the pharmacokinetics and exposure-response in the clinical pharmacology section of the label?	X			
General					
18	Are the clinical pharmacology and biopharmaceutics studies of appropriate design and breadth of investigation to meet basic requirements for approvability of this product?	X			
19	Was the translation (of study reports or other study information) from another language needed and provided in this submission?		X		

IS THE CLINICAL PHARMACOLOGY SECTION OF THE APPLICATION FILEABLE? Yes.

If the NDA/BLA is not fileable from the clinical pharmacology perspective, state the reasons and provide comments to be sent to the Applicant.

Please identify and list any potential review issues to be forwarded to the Applicant for the 74-day letter.

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CLINICAL PHARMACOLOGY AND BIOPHARMACEUTICS FILING FORM/CHECKLIST FOR NDA/BLA or Supplement

The sponsor will be asked to provide PK data in .xpt format from two studies. Data was only summarized in the electronic submission.

Kristina Estes	12 January 2012
Reviewing Clinical Pharmacologist	Date
Sue Chih Lee	17 January 2012
Team Leader/Supervisor	Date

File name: 5_Clinical Pharmacology and Biopharmaceutics Filing Form/Checklist for
NDA_BLA or Supplement 090808

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/s/

KRISTINA E ESTES
02/02/2012

SUE CHIH H LEE
02/03/2012

A. BIOPHARMACEUTICS FILING CHECKLIST				
	Parameter	Yes	No	Comment
1.	Does the application contain dissolution data?	X		
2.	Is the dissolution test part of the DP specifications?	X		
3.	Does the application contain the dissolution method development report?	X		Yes, but limited in scope.
4.	Is there a validation package for the analytical method and dissolution methodology?	X		
5.	Does the application include a biowaiver request?		X	
6.	Does the application include a IVIVC model?		X	
7.	Is information such as BCS classification mentioned, and supportive data provided?		X	
8.	Is information on mixing the product with foods or liquids included?		X	
9.	Is there any <i>in vivo</i> BA or BE information in the submission?	X		This part of the submission will be reviewed by the Office of Clinical Pharmacology.
B. filing conclusion				
	Parameter	Yes	No	Comment
10.	IS THE BIOPHARMACEUTICS SECTIONS OF THE APPLICATION FILEABLE?	X		
11.	If the NDA is not fileable from the product quality perspective, state the reasons and provide filing comments to be sent to the Applicant.	X		Fileable
12.	If the NDA is not fileable from the biopharmaceutics perspective, state the reasons and provide filing comments to be sent to the Applicant.			Fileable
13.	Are there any potential review issues to be forwarded to the Applicant for the 74-day letter?	X		See top of page 2

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/s/

MARK R SEGCEL
02/03/2012

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02/03/2012